```
from google.colab import files
import pandas as pd
import io
uploaded = files.upload()

filename = next(iter(uploaded))
df = pd.read_csv(io.BytesIO(uploaded[filename]))
df.head()
```

Choose Files sales.csv

sales.csv(text/csv) - 841 bytes, last modified: 10/27/2026 - 100% done

```
Saving sales.csv to sales (4).csv
```

	order_id	product	quantity	price	region
0	1001	Laptop	2.0	50000	South
1	1002	Mouse	5.0	500	North
2	1003	Keyboard	3.0	1500	East
3	1004	Monitor	2.0	12000	West
4	1002	Mouse	5.0	500	North

Next steps: Generate code with df | New interactive sheet

```
print("DataSet shape:",df.shape)
print("Column names:", df.columns.tolist())
df.info()
```

```
DataSet shape: (30, 5)
Column names: ['order_id', 'product', 'quantity', 'price', 'region']
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 5 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   order_id  30 non-null     int64
 1   product   30 non-null     object
 2   quantity  29 non-null     float64
 3   price     29 non-null     object
 4   region    29 non-null     object
dtypes: float64(1), int64(1), object(3)
memory usage: 1.3+ KB
```

Start coding or generate with AI.

```
missing_values = df.isnull().sum()
print("Missing values in each col", missing_values)
```

```
Missing values in each col order_id    0
product     0
quantity    1
price       1
region      1
dtype: int64
```

```
print("Number of duplicate records:", df.duplicated().sum())

# Display duplicate records
df[df.duplicated()]
```

```
Number of duplicate records: 1
```

	order_id	product	quantity	price	region
4	1002	Mouse	5.0	500	North

```
print("Number of duplicate order IDs:",
      df["order_id"].duplicated().sum())

df[df["order_id"].duplicated(keep=False)].sort_values("order_id")
```

```
Number of duplicate order IDs: 1
```

	order_id	product	quantity	price	region
1	1002	Mouse	5.0	500	North
4	1002	Mouse	5.0	500	North

```
df_clean = df.drop_duplicates(subset="order_id", keep="first")

print("Rows before removing duplicates:", len(df))
print("Rows after removing duplicates:", len(df_clean))

df_clean.head()
```

```
Rows before removing duplicates: 30
Rows after removing duplicates: 29
```

	order_id	product	quantity	price	region
0	1001	Laptop	2.0	50000	South
1	1002	Mouse	5.0	500	North
2	1003	Keyboard	3.0	1500	East
3	1004	Monitor	2.0	12000	West
5	1005	Phone	1.0	30000	South

Next steps: Generate code with `df_clean` New interactive sheet

```
# Convert price to numeric values
df_clean["price"] = pd.to_numeric(df_clean["price"], errors="coerce")

# Find the highest valid price
highest_price = df_clean["price"].max()

print("Highest product price:", highest_price)

# Display the corresponding order details
highest_price_order = df_clean[
    df_clean["price"] == highest_price
]

print("Order details for the highest-priced product:")
display(highest_price_order)
```

```
Highest product price: 50000.0
Order details for the highest-priced product:
/tmp/ipykernel_1029/174846231.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stab
  df_clean["price"] = pd.to_numeric(df_clean["price"], errors="coerce")
```

	order_id	product	quantity	price	region
0	1001	Laptop	2.0	50000.0	South
24	1024	Laptop	1.0	50000.0	East

Next steps: Generate code with `highest_price_order` New interactive sheet

```
# Convert quantity to numeric
df_clean["quantity"] = pd.to_numeric(
```

```
    df_clean["quantity"], errors="coerce"
)

# Calculate revenue for each order
df_clean["revenue"] = df_clean["quantity"] * df_clean["pr

# Calculate total revenue by region
regional_revenue = df_clean.groupby("region")["revenue"]

print("Total revenue by region:")
print(regional_revenue)
```

```
Total revenue by region:
region
East     213500.0
North     76500.0
South    238000.0
West      72000.0
Name: revenue, dtype: float64
/tmp/ipykernel_1029/942312589.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stab
  df_clean["quantity"] = pd.to_numeric(
/tmp/ipykernel_1029/942312589.py:7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stab
  df_clean["revenue"] = df_clean["quantity"] * df_clean["price"]
```

```
regional_revenue = (
    df_clean.groupby("region", as_index=False)["revenue"]
    .sum()
    .sort_values("revenue", ascending=False)
)

display(regional_revenue)
```

	region	revenue
2	South	238000.0
0	East	213500.0
1	North	76500.0
3	West	72000.0

Next steps: Generate code with regional_revenue | New interactive sheet